THE HIGHER DERIVED FUNCTORS OF THE ADEQUACY FUNCTOR VANISH ON MODULE-VALUED FUNCTORS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The functor Q and its adjointness to the inclusion of adequate functors. Same chapter, Lemma lemma-enough-injectives (whose notation the statement invokes), invoked at proof lines 1592-1594 and 1619-1620. Fixes Q and lets maps into Q(I^i) be transferred to maps into I^i.
- The two-term spectral sequence for the derived functors of a composition. Derived Categories, Lemma derived-lemma-two-ss-complex-functor, invoked at proof lines 1601-1606. Reduces the claim for F to the claim for a module-valued functor.
- Every adequate functor is a quotient of a direct sum of linearly adequate functors. Same chapter, Lemma lemma-adequate-surjection-from-linear, invoked at proof lines 1614-1618. Produces the linearly adequate test functor L in the contradiction.
- The vanishing of positive Ext groups out of a linearly adequate functor. Same chapter, Lemma lemma-ext-group-zero, invoked at proof lines 1621-1623. Contradicts the constructed nonzero Ext class.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06UX: sources/stacks-adequate/pinned-original-source.tex, source lines 468–468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06Z7: sources/stacks-adequate/pinned-original-source.tex, source lines 1086–1086. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 06ZH: sources/stacks-adequate/pinned-original-source.tex, source lines 1557–1557. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *With notation as in Lemma 06Z7 we have $R^pQ(F) = 0$ for all $p > 0$ and any adequate functor F.*

Proof. Choose an exact sequence $0 \to F \to \underline{M^0} \to \underline{M^1}$. Set $M^2 = \mathrm{Coker}(M^0 \to M^1)$ so that $0 \to F \to \underline{M^0} \to \underline{M^1} \to \underline{M^2} \to 0$ is a resolution. By Derived Categories, Lemma 015J we obtain a spectral sequence

$$R^pQ(\underline{M^q}) \Rightarrow R^{p+q}Q(F)$$

Since $Q(\underline{M^q}) = \underline{M^q}$ it suffices to prove $R^pQ(\underline{M}) = 0$, $p > 0$ for any A-module M.

Choose an injective resolution $\underline{M} \to I^\bullet$ in the category $\mathcal{P}$. Suppose that $R^iQ(\underline{M})$ is nonzero. Then $\mathrm{Ker}(Q(I^i) \to Q(I^{i+1}))$ is strictly bigger than the image of $Q(I^{i-1}) \to Q(I^i)$. Hence by Lemma 06UX there exists a linearly adequate functor L and a map $\varphi : L \to Q(I^i)$ mapping into the kernel of $Q(I^i) \to Q(I^{i+1})$ which does not factor through the image of $Q(I^{i-1}) \to Q(I^i)$. Because Q is a left adjoint to the inclusion functor the map φ corresponds to a map $\varphi' : L \to I^i$ with the same properties. Thus φ' gives a nonzero element of $\mathrm{Ext}^i_{\mathcal{P}}(L, \underline{M})$ contradicting Lemma 06ZH. $\square$